How-To Library

Informatica

How to Migrate Microsoft SQL Server Connections from the OLE DB to the ODBC Provider Type

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Abstract

When you create a Microsoft SQL Server connection, you can use the OLE DB or ODBC provider types. If required, you can migrate the OLE DB provider type to the ODBC provider type. This article explains how to migrate Microsoft SQL Server connections from the OLE DB provider type to the ODBC provider type.

Supported Versions

- PowerCenter 10.2
- Informatica Platform 10.2

Table of Contents

Overview. 2
Questions About Migrating the Microsoft SQL Server Provider Type. 3
Are there data type support changes between the ODBC and OLE DB provider types?. 3
What are the rules and guidelines that I must consider when I use the ODBC provider type?. 5
Are there any known limitations with using the ODBC provider type?. 6
Can I migrate Microsoft SQL Server connections from the OLE DB to the ODBC provider type?. 6
Do I have to recreate the mappings?. 7
After I migrate to the ODBC provider type, what are the post-migration tasks that I must perform to retain the previous behavior with the OLE DB provider type?. 7
Questions About Creating Microsoft SQL Server Connections. 7
Which provider type should I use when I create a new Microsoft SQL Server connection?. 7
Should I create a new Microsoft SQL Server connection by using the DSN or without using the DSN?. 8
Creating a Microsoft SQL Server connection in PowerCenter. 8
How to create a SQL Server connection in Informatica Developer?. 10
How can I create a SQL Server connection in Informatica Analyst? 12
How can I create a Microsoft SQL Server connection for PowerCenter by using the Informatica command line program?. 14
How to create a MS SQL Server connection for Informatica Developer by using the Informatica command line program?. 14

Overview

You can configure native connectivity to a Microsoft SQL Server database by using the OLE DB or ODBC provider types.

If required, you can migrate the OLE DB provider type to the ODBC provider type. Informatica provides commands to easily migrate Microsoft SQL Server connections from the OLE DB provider type to the ODBC provider type.

Note: You must use the DataDirect ODBC driver to connect to the Microsoft SQL Server database with the ODBC provider type. Informatica does not support other third-party ODBC drivers.

Questions About Migrating the Microsoft SQL Server Provider Type

Are there data type support changes between the ODBC and OLE DB provider types?

No. Both the ODBC and OLE DB provider types support the following data types:

Microsoft SQL Server Data Type	Range	Transformation Data Type	Range
Binary	1 to 8,000 bytes	Binary	1 to 104,857,600 bytes
bigint	- 9,223,372,036,854,775,808 to 9,223,372,036,854,775,807	bigint	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 Precision 19, scale 0
Bit	1 bit	String	1 to 104,857,600 characters
Char	1 to 8,000 characters	String	1 to 104,857,600 characters
Datetime	Jan 1, 1753 A.D. to Dec 31, 9999 A.D. Precision 23, scale 3 (precision to 3.33 milliseconds)	Date/Time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. (precision to the nanosecond)
Datetime2	Jan 1, 0001 A.D. 00:00:00 to Dec 31, 9999 A.D. 23:59:59.9999999	Timestamp	Precision 22 to 27
Decimal	Precision 1 to 38, scale 0 to 38	Decimal	PowerCenter: Precision 1 to 28, scale 0 to 28 Informatica Developer: For transformations that support precision up to 38 digits, the precision is 1 to 38 digits, and the scale is 0 to 38. If you specify the precision greater than the maximum number of digits, the Data Integration Service converts decimal values to double in high precision mode.
Float	-1.79E+308 to 1.79E+308	Double	Precision 15
Image	1 to 2,147,483,647 bytes	Binary	1 to 104,857,600 bytes
Int	-2,147,483,648 to 2,147,483,647	Integer	-2,147,483,648 to 2,147,483,647 Precision 10, scale 0
Money	-922,337,203,685,477.5807 to 922,337,203,685,477.5807	Decimal	Precision 1 to 28, scale 0 to 28

Microsoft SQL Server Data Type	Range	Transformation Data Type	Range
nchar	1 to 4000 characters	String	1 to 104,857,600 characters
ntext	1 to 1,073,741,823 bytes	Text	1 to 104,857,600 characters
Numeric	Precision 1 to 38, scale 0 to 38	Decimal	PowerCenter: Precision 1 to 28, scale 0 to 28 Informatica Developer: For transformations that support precision up to 38 digits, the precision is 1 to 38 digits, and the scale is 0 to 38. For transformations that support precision up to 28 digits, the precision is 1 to 28 digits, and the scale is 0 to 28. If you specify the precision greater than the maximum number of digits, the Data Integration Service converts decimal values to double in high precision mode.
nvarchar	1 to 4000 characters	String	1 to 104,857,600 characters
Real	-3.40E+38 to 3.40E+38	Double	Precision 15
Smalldatetime	Jan 1, 1900, to June 6, 2079 Precision 19, scale 0 (precision to the minute)	Date/Time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. (precision to the nanosecond)
Smallint	-32,768 to 32,768	Integer	-2,147,483,648 to 2,147,483,647 Precision 10, scale 0
Smallmoney	-214,748.3648 to 214,748.3647	Decimal	Precision 1 to 28, scale 0 to 28
Sysname	1 to 128 characters	String	1 to 104,857,600 characters
Text	1 to 2,147,483,647 characters	Text	1 to 104,857,600 characters
Timestamp	8 bytes	Binary	1 to 104,857,600 bytes
Tinyint	0 to 255	Integer	-2,147,483,648 to 2,147,483,647 Precision 10, scale 0
Uniqueidentifier	Precision 38, scale 0	String	To successfully move or change Uniqueidentifier data, ensure that the data is in the following format: `xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx` where `x` is a hexadecimal digit in the range 0-9 or a-f.

Microsoft SQL Server Data Type	Range	Transformation Data Type	Range
Varbinary	1 to 8,000 bytes	Binary	1 to 104,857,600 bytes
Varchar	1 to 8,000 characters	String	1 to 104,857,600 characters

The following table compares Microsoft SQL Server synonyms to PowerCenter transformation data types:

Synonym	Transformation
Binary Varying	Binary
Character	String
Character Varying	String
Dec	Decimal
Double Precision	Double
Integer	Integer

Uniqueidentifier Data Type

Uniqueidentifier is a Microsoft SQL Server data type that is used to store Globally Unique Identifiers (GUIDs). It can store 16 bytes of data.

PowerCenter imports the Microsoft SQL Server Uniqueidentifier data type as a Microsoft SQL Server Varchar data type of 38 characters.

The Developer tool treats the Uniqueidentifier data type as String. To move or change Uniqueidentifier data, connect the Uniqueidentifier column to a String column. To successfully move or change Uniqueidentifier data, ensure that the data is in the following format:

```
xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx
```

where `x` is a hexadecimal digit in the range 0-9 or a-f.

For example, `6F9619FF-8B86-D011-B42D-00C04FC964FF` is a valid Uniqueidentifier value.

The Developer tool can store 16 bytes of Uniqueidentifier data in 36 characters. However, since Uniqueidentifier data can be represented within two curly brackets, the Developer tool assigns two additional characters to the precision to accommodate the curly brackets. When you connect a Uniqueidentifier column to a String column, set the precision of the String column to 38 to successfully move or change Uniqueidentifier data.

What are the rules and guidelines that I must consider when I use the ODBC provider type?

Consider the following rules and guidelines when you configure ODBC connectivity to a Microsoft SQL Server database:

- If you want to use a Microsoft SQL Server connection without using a Data Source Name (DSN less connection), you must configure the odbcinst.ini environment variable.
- If you use a DSN connection, you must add the entry EnableQuotedIdentifiers=1 to the ODBC DSN. If you do not add the entry, data preview and mapping run fail.

- You can use the Microsoft SQL Server NTLM authentication on a DSN less Microsoft SQL Server connection on Microsoft Windows.
- You cannot use an SSL connection on a DSN less connection. If you want to use SSL, you must use the DSN connection. Enable the **Use DSN** option and configure the SSL options in the odbc.ini file.
- If the Microsoft SQL Server uses Kerberos authentication, you must set the GSSClient property to point to the Informatica Kerberos libraries. Use the following path and file name:
  `<Informatica installation directory>/server/bin/libgssapi_krb5.so.2`

  Create an entry for the GSSClient property in the DSN entries section in odbc.ini for a DSN connection or in the SQL Server wire protocol section in odbcinst.ini for a connection that does not use DSN.

Are there any known limitations with using the ODBC provider type?

The following known limitations apply when you use the ODBC provider type:

- The data format for the UUID data type is inconsistent between the ODBC and OLE DB provider types.
- When there is an error record, Microsoft SQL Server bulk load sessions display incorrect statistics in the session log.
- If you override the SQL query, and use a SELECT statement with a datetime literal or NULL constant value, the Microsoft SQL session fails at run time.

Can I migrate Microsoft SQL Server connections from the OLE DB to the ODBC provider type?

Yes, if you had created a Microsoft SQL Server connection in a version earlier than 10.0, you can easily migrate the connection to use the ODBC provider type. You can migrate connections that you created for both PowerCenter and Informatica Developer. You do not have to recreate the mappings after you migrate the connections. After you upgrade to Informatica version 10.0 or later, all Microsoft SQL Server connections are set to the OLE DB provider type by default.

You can upgrade all the Microsoft SQL Server connections to the ODBC provider type by running the following commands:

- If you use PowerCenter, run the following command:
  pmrep upgradeSqlServerConnection
- If you use Informatica Developer, run the following command:
  infacmd.sh isp upgradeSQLSConnection

On Linux, after you run the upgrade commands, you must set the **ODBCINST** environment variable as follows:

ODBCINST=`<Informatica installation directory>/ODBC7.1/odbcinst.ini`

You must set the **ODBCINST** environment variable on each machine that hosts the Informatica services. After you set the environment variable, you must restart the node that hosts the Informatica services for the changes to take effect.

Do I have to recreate the mappings?

No, you do not have to recreate the mappings. You can use the same mappings after you upgrade the Microsoft SQL Server connections from the OLE DB to the ODBC provider type.

After I migrate to the ODBC provider type, what are the post-migration tasks that I must perform to retain the previous behavior with the OLE DB provider type?

The following table describes solutions for issues that you might encounter after you migrate to the ODBC provider type:

Issue	Resolution
PowerCenter sessions write data of the Bit data type as 1 and 0 to Microsoft SQL Server targets.	Set the value of the ProcesSQLServerBITASVarchar custom property to **Yes** to write the Bit data as T and F instead of 1 and 0.
PowerCenter sessions that write data of the BIT data type to Microsoft SQL Server targets fail if all of the following conditions are true: - You configure the Microsoft SQL Server connection to use the ODBC provider type. - You configure the data movement mode for the PowerCenter Integration Service as ASCII. - You configure the value of the custom property ProcesSQLServerBITASVarchar as Yes.	Set the value of the ProcesSQLServerBITASVarchar custom property to **No** or set the data movement mode to **Unicode**.
PowerCenter sessions that write data of the Smalldatetime data type to Microsoft SQL Server targets fail with a data overflow error.	Set the value of the SQLServerLegacySmallDateTime custom property to **Yes**.
PowerCenter sessions that write data of the Date data type to Microsoft SQL Server targets fail.	Set the value of the SQLServerBulkSupportDateFormat custom property to **YYYYMMDD** to support the MM/DD/YYYY and YYYYMMDD date formats.

Questions About Creating Microsoft SQL Server Connections

Which provider type should I use when I create a new Microsoft SQL Server connection?

In Informatica version 10.0 or later, when you create a new Microsoft SQL Server connection, the provider type is set to ODBC by default. If required, you can change the provider type to OLE DB.

If you choose the ODBC provider type, you can enable the **Use DSN** option to use the DSN configured in the Microsoft ODBC Administrator on Windows or DSN configured in the `odbc.ini` file on UNIX.

If you choose the OLE DB provider type, you must install the Microsoft SQL Server Native Client to configure native connectivity to the Microsoft SQL Server database.

Should I create a new Microsoft SQL Server connection by using the DSN or without using the DSN?

You can create a Microsoft SQL Server connection either by using the DSN or without using the DSN.

You can create a connection by using the DSN if you want to specify ODBC driver properties such as AnsiNPW, AuthenticationMethod, and GSSClient. To configure SSL, you must use the DSN option.

If you do not enable the **Use DSN** option, you must specify the server name and database name in the connection properties. You can create a connection without using the DSN if you do not want to specify any ODBC driver property.

Creating a Microsoft SQL Server connection in PowerCenter

Perform the following steps to create a Microsoft SQL Server connection in PowerCenter:

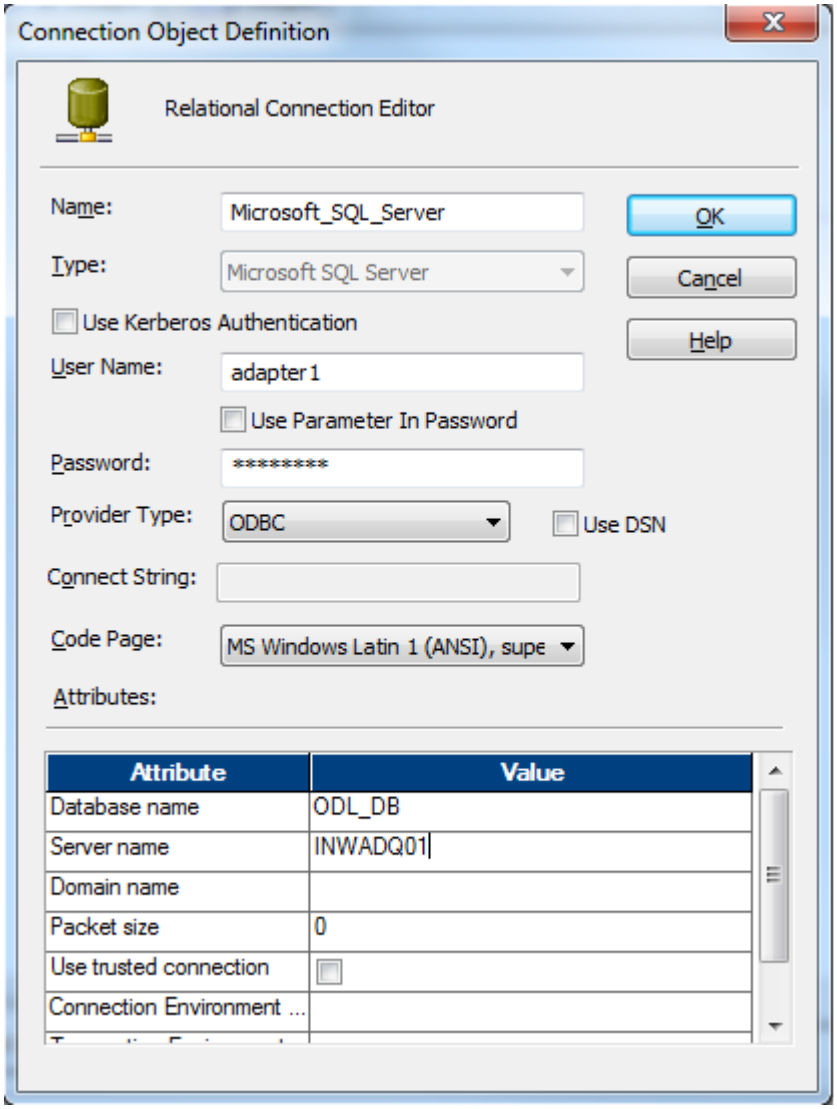
1. In the PowerCenter Workflow Manager, click **Connections** > **Relational**.

   The **Connection Browser** dialog box appears.
2. Click **New**.

   The **Select Subtype** dialog box appears.
3. Select the type as **Microsoft SQL Server** and click **OK**.

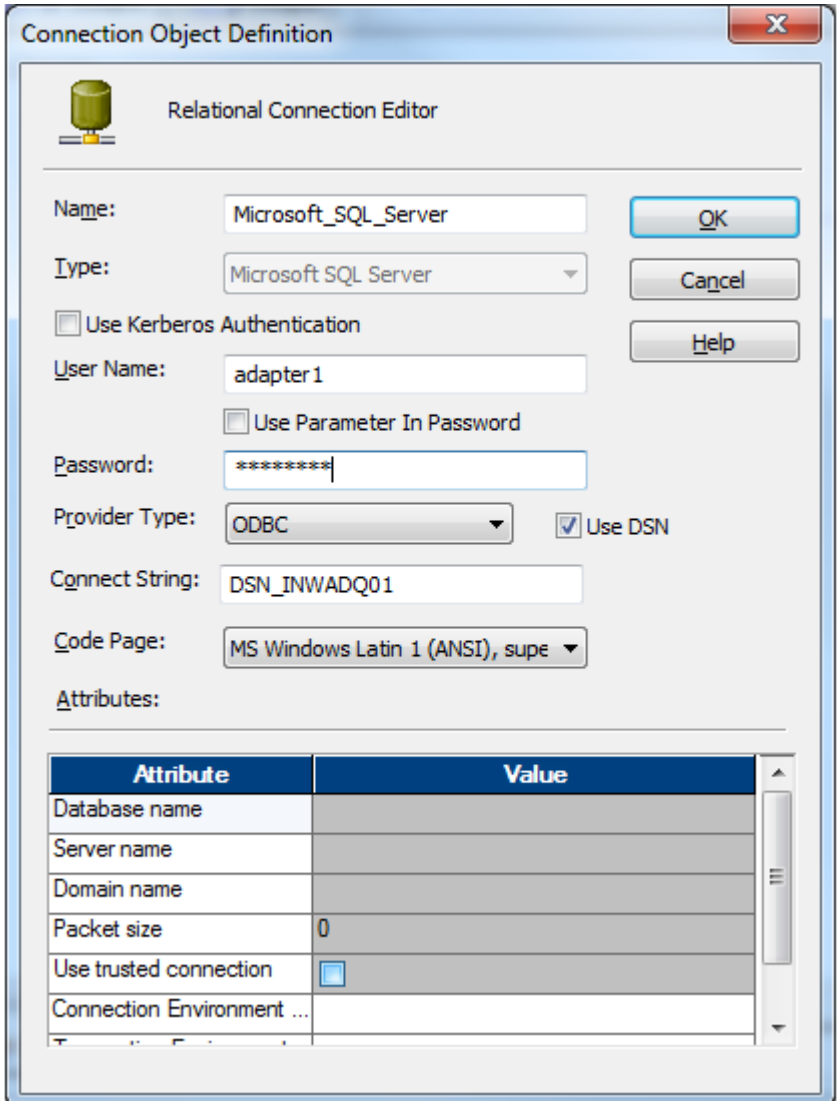
   The **Relational Connection Editor** dialog box appears.
4. Enter a connection name.
5. Enter the user name and password to access the Microsoft SQL Server database.
6. Select the provider type as **ODBC**.
7. Perform one of the following tasks:
   - **DSN-less Connection:** By default, the **Use DSN** check box is not selected. The **Connect String** field is disabled and the **Attribute** fields are enabled. Enter the database name, server name, and other connection attributes in the **Attribute** section.

The following image shows the **Connection Object Definition** dialog box when the **Use DSN** option is not selected:



- **DSN Connection:** Select the **Use DSN** check box to create a connection by using the DSN. The **Connect String** field is enabled and the **Attribute** fields are disabled. Enter the DSN name as the connect string.

The following image shows the **Connection Object Definition** dialog box when the **Use DSN** option is selected:



8. Configure the other connection properties.
9. Click **OK** to create the connection.

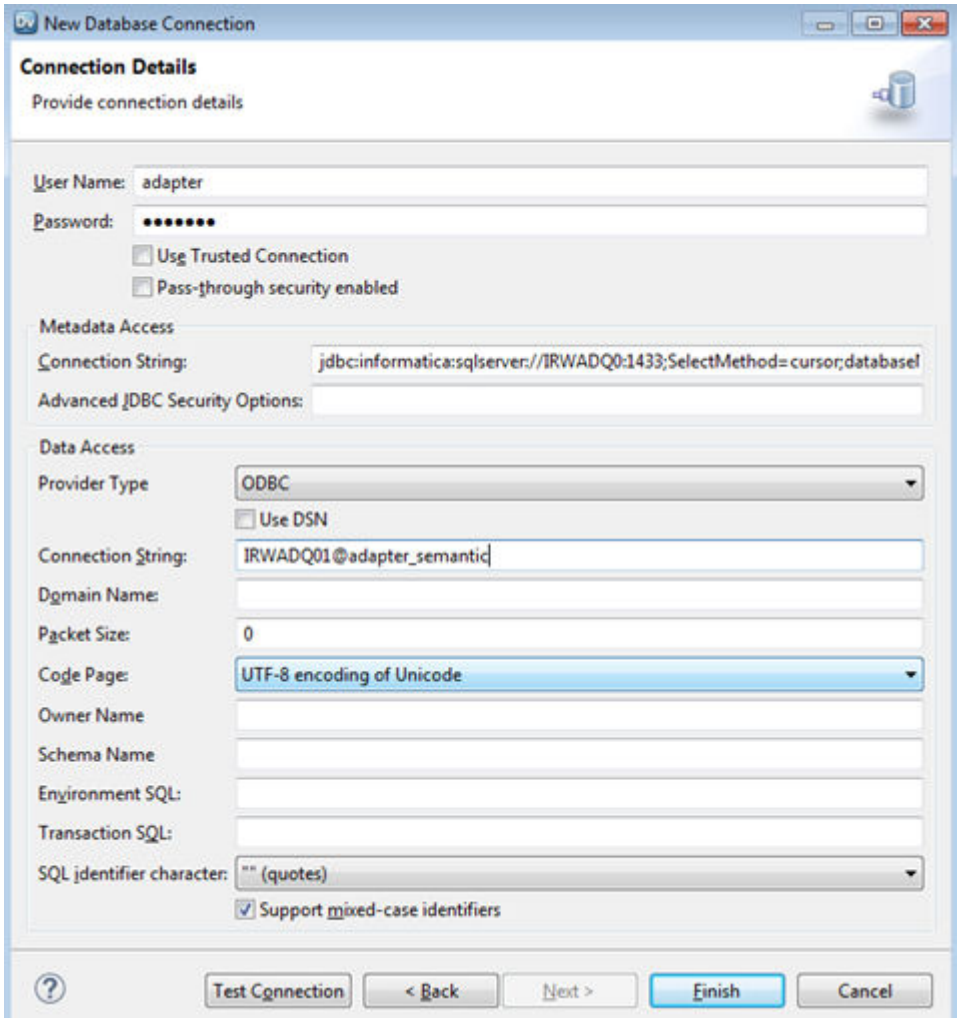
How to create a SQL Server connection in Informatica Developer?

Perform the following steps to create a Microsoft SQL Server connection in Informatica Developer:

1. In Informatica Developer, click **Window** > **Preferences**.
2. Select **Informatica** > **Connections**.
3. Expand the domain in the **Available Connections** list.
4. Select **Database** > **Microsoft SQL Server** > **Add**.

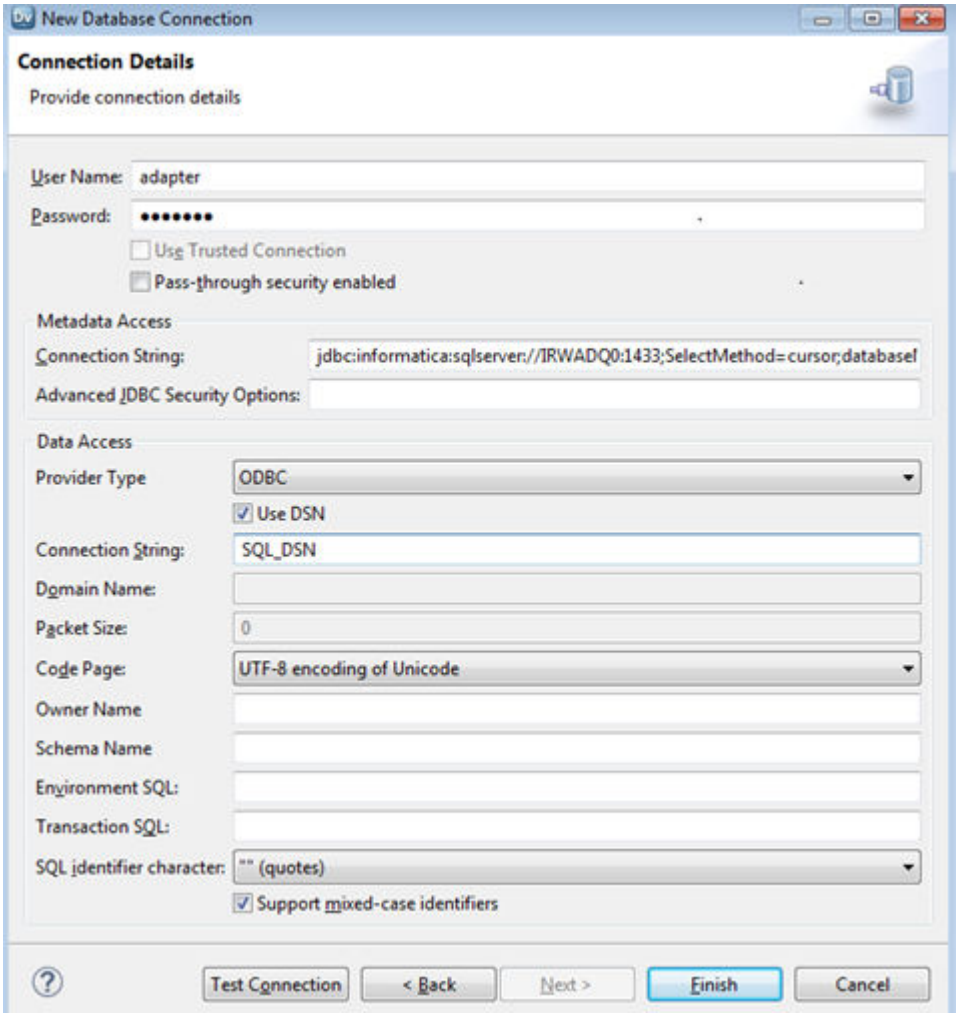
   The **New Database Connection** dialog box appears.
5. Enter the connection name and click **Next**.
6. Enter the user name and password to access the Microsoft SQL Server database.

7. Enter the Microsoft SQL Server database host name, port number, and database name in the connection string for metadata access.
8. Select the provider type as **ODBC** for data access.
9. Perform one of the following tasks:
   - **DSN-less Connection:** By default, the **Use DSN** check box is not selected. Enter the connection string in the following format: <servername>@<dbname>
     The following image shows the **New Database Connection** dialog box when the **Use DSN** option is not selected:



   - **DSN Connection:** Select the **Use DSN** check box to create a connection by using the DSN. The **Use Trusted Connection** and some other connection attributes are disabled. Enter the DSN name in the connection string.

The following image shows the **New Database Connection** dialog box when the **Use DSN** option is selected:



10. Configure the other connection properties.
11. Click **Finish** to create the connection.

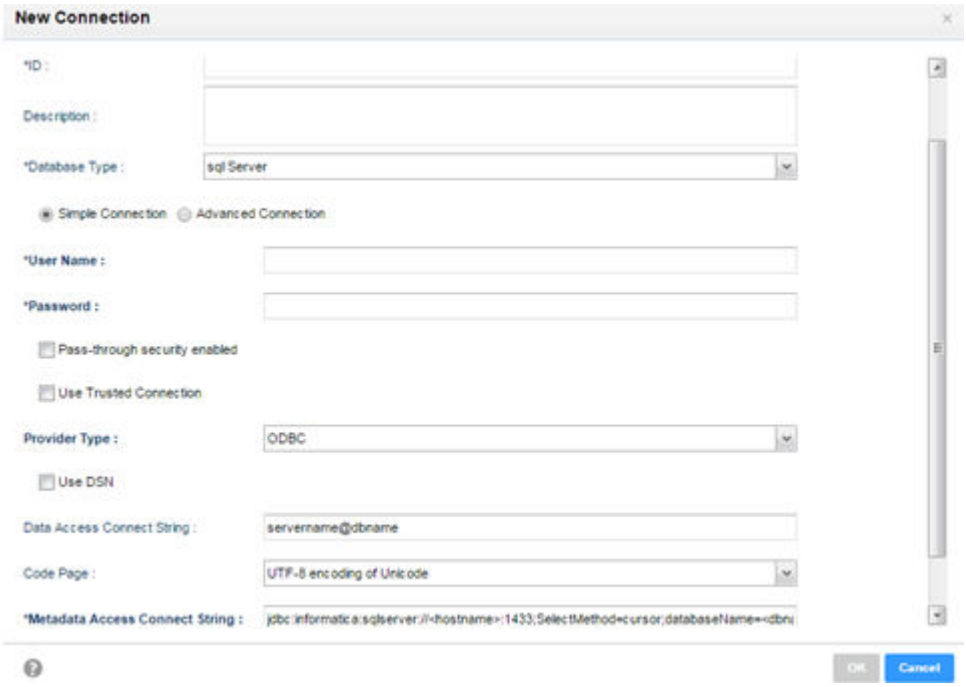
How can I create a SQL Server connection in Informatica Analyst?

Perform the following steps to create a Microsoft SQL Server connection in Informatica Analyst:

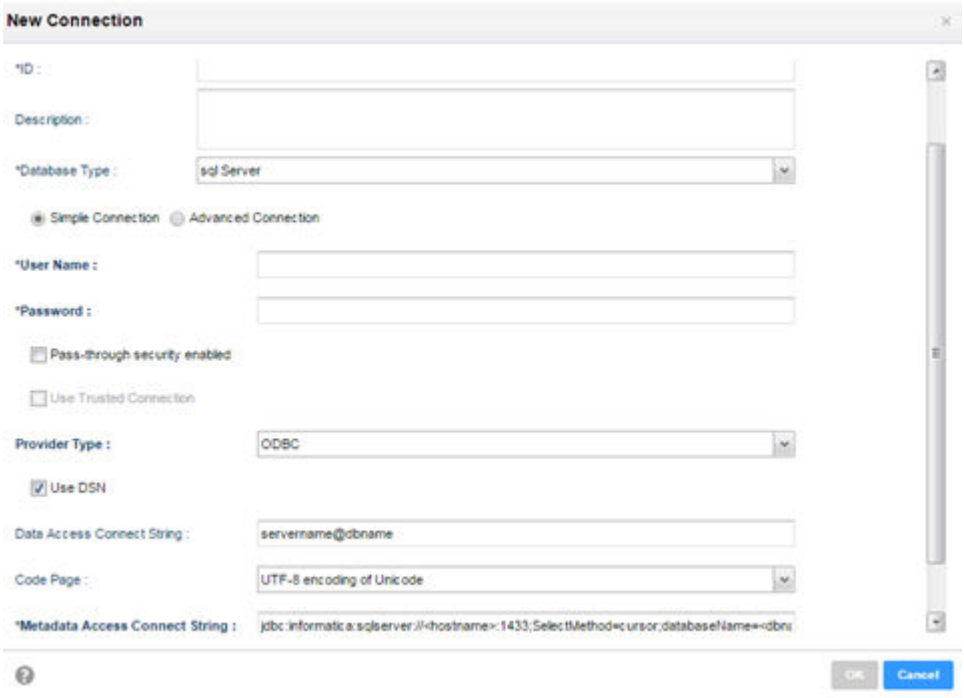
1. In Informatica Analyst, click **Manager** > **Connections**.
2. Click the **New** icon.

   The **New Connection** dialog box appears.
3. Select the database type as **sql Server** from the menu.
4. Enter the user name and password to access the Microsoft SQL Server database.
5. Select the provider type as **ODBC** for data access.
6. Perform one of the following tasks:

- **DSN-less Connection:** By default, the **Use DSN** check box is not selected. Enter the data access connect string in the following format: <servername>@<dbname>
  The following image shows the **New Database Connection** dialog box when the **Use DSN** option is not selected:



- **DSN Connection:** Select the **Use DSN** check box to create a connection by using the DSN. Enter the DSN name in the data access connect string.
  The following image shows the **New Database Connection** dialog box when the **Use DSN** option is selected:



7. Enter the Microsoft SQL Server database host name, port number, and database name in the **Metadata Access Connect String**. Enter the metadata access connect string in the following format: jdbc:informatica:sqlserver://<hostname>:<portno>;databaseName=<database_name>
8. Click **OK** to create the connection.

How can I create a Microsoft SQL Server connection for PowerCenter by using the Informatica command line program?

Run the pmrep CreateConnection command to create a Microsoft SQL Server connection for PowerCenter.

The pmrep CreateConnection command uses the following syntax:

```
-r <repo>
-s <connection_type>
-n <connection_name>
[{-u <user_name>
[-p <password> |
-P <password_environment_variable>]}|
-K <connection_to_the_Kerberos_server>]
[-c <connect_string> (required for Oracle, Informix, DB2, Microsoft SQL Server, ODBC, and
NetezzaRelational)]
[-l <code_page>]
[-r <rollback_segment> (valid for Oracle connection only)]
[-e <connection_environment_SQL>]
[-f <transaction_environment_SQL>]
[-z <packet_size> (valid for Sybase ASE and MS SQL Server connection)]
[-b <database_name> (valid for Sybase ASE, Teradata and MS SQL Server connection)]
[-v <server_name> (valid for Sybase ASE and MS SQL Server connection)]
[-d <domain_name> (valid for MS SQL Server connection only)]
[-t (enable trusted connection, valid for MS SQL Server connection only)]
[-x (enable advanced security, lets users give Read, Write and Execute permissions only for
themselves.)]
[-k <connection_attributes> (attributes have the format name=value;name=value; and so on)]
[-y (Provider Type (1 for ODBC and 2 for OLEDB), valid for MS SQL Server connection only)]
[-m (UseDSN, valid for MS SQL Server connection only)]
```

Enter the following options to specify the provider type values for a Microsoft SQL Server connection:

-y: This argument specifies the provider type. You can specify the following values:

- Set the value to 1 if you want to use the ODBC provider type. Default is 1.
- Set the value to 2 if you want to use the OLE DB provider type.

-m: This argument enables the **Use DSN** attribute when you use the ODBC provider type. The Integration Service retrieves the database and server names from the DSN.

-c: This argument specifies the connect string.

Note: You must specify the -c argument if you use the -m argument.

How to create a MS SQL Server connection for Informatica Developer by using the Informatica command line program?

Run the infacmd isp CreateConnection command to create a Microsoft SQL Server connection for Informatica Developer.

The infacmd isp CreateConnection command uses the following syntax:

```
<-DomainName|-dn> domain_name
<-UserName|-un> user_name
<-Password|-pd> password
[<-SecurityDomain|-sdn> security_domain]
```

```
[<-ResilienceTimeout|-re> timeout_period_in_seconds]
<-ConnectionName|-cn> connection_name
[<-ConnectionId|-cid> connection_id]
<-ConnectionType|-ct> connection_type
[<-ConnectionUserName|-cun> connection_user_name]
[<-ConnectionPassword|-cpd> connection_password]
[<-VendorId|-vid> vendor_id]
[-o options] (name-value pairs separated by space)
```

Enter the following options to specify the provider type values for a Microsoft SQL Server connection:

- Set the value to **ProviderType=0** to use the ODBC provide type. Default is 0.
- Set the value to **ProviderType=1** to use the OLE DB provide type.
- Set the value to **UseDsn=true** to enable the DSN.
- Set the value to **UseDsn=false** to disable the DSN. By default, DSN is disabled.

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